Chapter 2

Classification of Materials: Intrinsic vs. Extrinsic Semiconductors

2.1 Introduction

In Lecture 1, we explored the concept of **Intrinsic Semiconductors**—pure crystals of Silicon or Germanium. We learned that at room temperature, their conductivity is limited because only a small number of electron-hole pairs are generated thermally.

For practical electronic devices like diodes and transistors, pure Silicon is actually a poor conductor. To make it useful, we must significantly increase its conductivity. We achieve this not by heating it, but by altering its chemical composition through a process called **Doping**.

Learning Objectives

By the end of this chapter, you should be able to:

- Define Doping and explain why it is necessary in electronics.
- Distinguish between N-Type and P-Type semiconductors.
- Draw the crystal lattice structures showing Donor and Acceptor atoms.
- Analyze Energy Band diagrams to locate Donor and Acceptor energy levels.
- Apply the Mass Action Law to calculate carrier concentrations.

2.2 The Doping Process

Doping is the deliberate addition of a specific impurity element to a pure (intrinsic) semiconductor to alter its electrical properties.

2.2.1 Why Dope?

In intrinsic Silicon at room temperature (300 K), the intrinsic carrier concentration (n_i) is approximately 1.5×10^{10} electrons/cm^3. While this sounds like a lot, compared to a metal like copper (10^{22} electrons/cm^3), it is extremely low.

By adding just **one** impurity atom for every 10^6 to 10^8 Silicon atoms, we can increase the conductivity by a factor of thousands. This creates an **Extrinsic Semiconductor**.

2.2.2 Types of Impurities

The impurities (dopants) are chosen based on the number of valence electrons they possess relative to Silicon (which has 4).

1. **Pentavalent Impurities (Group V):** Atoms with **5** valence electrons.
   - Examples: Phosphorus (P), Arsenic (As), Antimony (Sb).
   - Used to create **N-Type** material.

2. **Trivalent Impurities (Group III):** Atoms with **3** valence electrons.
   - Examples: Boron (B), Gallium (Ga), Indium (In).
   - Used to create **P-Type** material.

2.3 N-Type Semiconductors

An N-Type semiconductor is created by doping pure Silicon with a Pentavalent impurity. The "N" stands for **Negative**, referring to the charge of the majority carriers (electrons).

2.3.1 Atomic Structure

Let us visualize what happens when an Arsenic atom (5 valence electrons) is inserted into a Silicon lattice (4 valence electrons):

1. The Arsenic atom displaces a Silicon atom in the crystal structure.
2. Four of the Arsenic's valence electrons form covalent bonds with the four neighboring Silicon atoms.
3. The **fifth electron** has no "partner" to bond with. It remains loosely attached to the parent atom.
4. Because it is not part of a bond, very little energy (room temperature heat is enough) is required to detach this fifth electron.
5. It becomes a **Free Electron** in the conduction band.

Since each impurity atom "donates" a free electron to the conduction band, these impurities are called **Donor Atoms**.

2.3.2 Majority and Minority Carriers

In an N-Type material:

- **Majority Carriers:** Electrons. (Provided by the doping process).

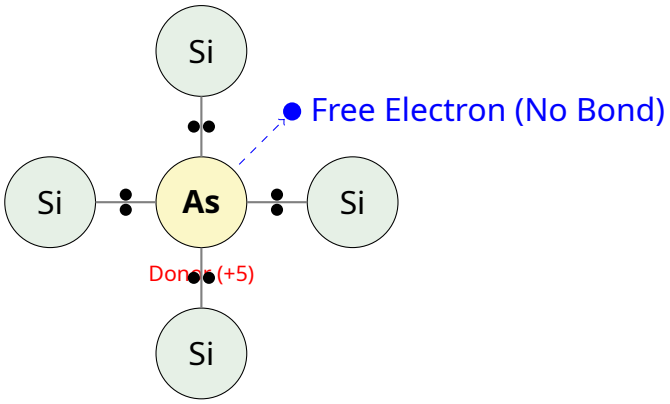


Figure 2.1: Crystal Lattice of N-Type Silicon showing the extra free electron from the Donor atom.

- **Minority Carriers:** Holes. (Generated only by thermal agitation breaking Si-Si bonds).

Mathematically: $n \gg p$.

2.4 P-Type Semiconductors

A P-Type semiconductor is created by doping pure Silicon with a Trivalent impurity. The ”P” stands for **Positive**, referring to the majority carriers (holes).

2.4.1 Atomic Structure

When a Boron atom (3 valence electrons) enters the lattice:

1. It forms covalent bonds with three neighboring Silicon atoms.
2. The fourth bond is incomplete because Boron has no electron to share there.
3. This ”missing electron” or vacancy is called a **Hole**.
4. This hole is ready to accept a free electron from a neighboring atom.

Since each impurity atom creates a hole that can ”accept” an electron, these impurities are called **Acceptor Atoms**.

2.4.2 Majority and Minority Carriers

In a P-Type material:

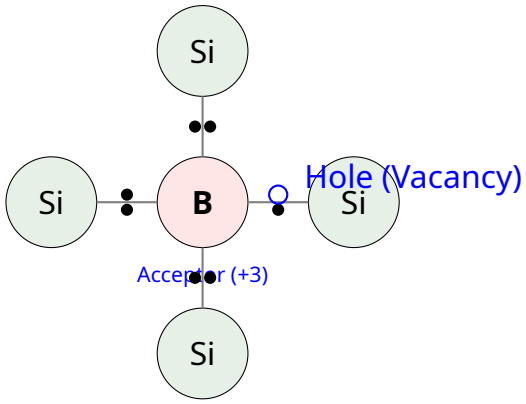


Figure 2.2: Crystal Lattice of P-Type Silicon showing the Hole created by the Acceptor atom.

- **Majority Carriers:** Holes. (Provided by doping).
- **Minority Carriers:** Electrons. (Generated thermally).

Mathematically: $p \gg n$.

2.5 Energy Band Diagrams

Doping introduces new energy levels within the forbidden gap (E_g).

2.5.1 Donor Level (N-Type)

In N-type material, the extra electrons are very loosely bound. They occupy an energy level called the **Donor Level (E_D)**, which sits *just below* the Conduction Band. It takes very little energy (≈ 0.05 eV) for these electrons to jump into the Conduction Band.

2.5.2 Acceptor Level (P-Type)

In P-type material, the holes are ready to accept electrons. This creates an **Acceptor Level (E_A)** located *just above* the Valence Band. Valence electrons can easily jump into this level, leaving holes behind in the Valence Band.

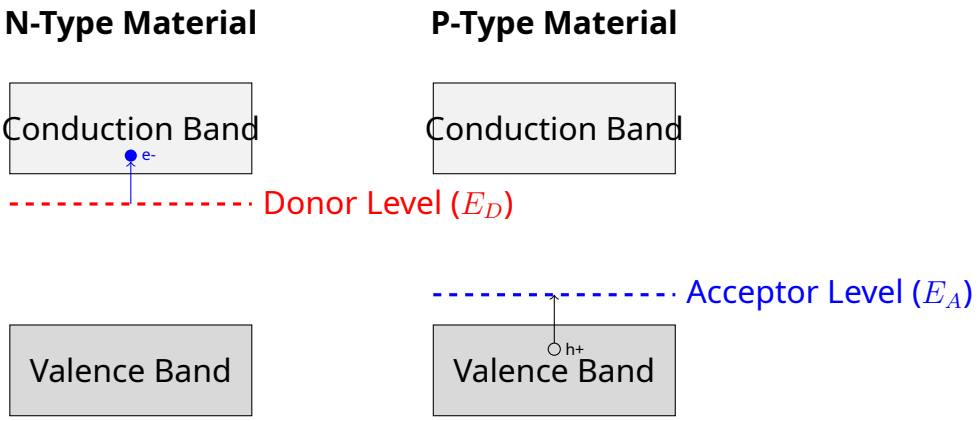


Figure 2.3: Energy Band diagrams showing the position of Donor and Acceptor energy levels.

2.6 The Fermi Level

The **Fermi Level (E_F)** is defined as the energy state that has a 50% probability of being occupied by an electron.

- **Intrinsic:** E_F is exactly in the middle of the forbidden gap.
- **N-Type:** Because there are many electrons, E_F shifts **upward**, closer to the Conduction Band.
- **P-Type:** Because there are many holes (vacancies), E_F shifts **downward**, closer to the Valence Band.

2.7 Mass Action Law

Under thermal equilibrium, there is a fundamental relationship between the concentration of free electrons (n) and holes (p).

Law of Mass Action

The product of the free electron concentration and the hole concentration is constant at a given temperature.

$$n \cdot p = n_i^2 \tag{2.1}$$

Where n_i is the intrinsic carrier concentration.

This law implies that as we increase the number of majority carriers (via doping), the number of minority carriers decreases proportionally.

Example 1: Calculation

Pure Silicon has $n_i = 1.5 \times 10^{10}$ cm^{-3}. It is doped with 10^{15} Donor atoms/cm^3. Calculate the concentration of minority carriers (holes).

Solution: Since $N_D \gg n_i$, we approximate $n \approx N_D = 10^{15}$. Using Mass Action Law:

$$p = \frac{n_i^2}{n} = \frac{(1.5 \times 10^{10})^2}{10^{15}}$$

$$p = \frac{2.25 \times 10^{20}}{10^{15}} = 2.25 \times 10^5 \text{ holes/cm}^3$$

2.8 Charge Neutrality

A very common misconception is that N-type material is negatively charged and P-type is positively charged. **This is FALSE.**

Extrinsic semiconductors are electrically NEUTRAL.

- **N-Type:** While we added free electrons (negative), each electron came from a neutral Donor atom. When the electron leaves, the Donor atom becomes a **Positive Ion**.

$$\text{Total Negative Charge (Electrons)} = \text{Total Positive Charge (Ions)}$$

- **P-Type:** While we added holes, each hole corresponds to an Acceptor atom. When the Acceptor takes an electron, it becomes a **Negative Ion**.

2.9 Review Questions

1. Explain the difference between a Donor impurity and an Acceptor impurity.
2. Why does the Fermi level shift upwards in an N-type semiconductor?
3. A silicon wafer is doped with Arsenic. Is the resulting material N-type or P-type? Is the material positively charged, negatively charged, or neutral? Explain.

4. Describe the position of the Donor Energy Level (E_D) relative to the Conduction Band.

5. Using the Mass Action Law, explain what happens to the hole concentration if we double the doping concentration of donor atoms.